

The background is a gradient of dark blue and purple. On the left side, there are several concentric circles and a large arc with a scale from 140 to 260. The scale is marked with numbers every 10 units. There are also some smaller circles and arrows scattered across the background.

NATIONWIDE AUCTION SYSTEM

BUSINESS PRESENTATION

YINA CAI

KATA OVERVIEW

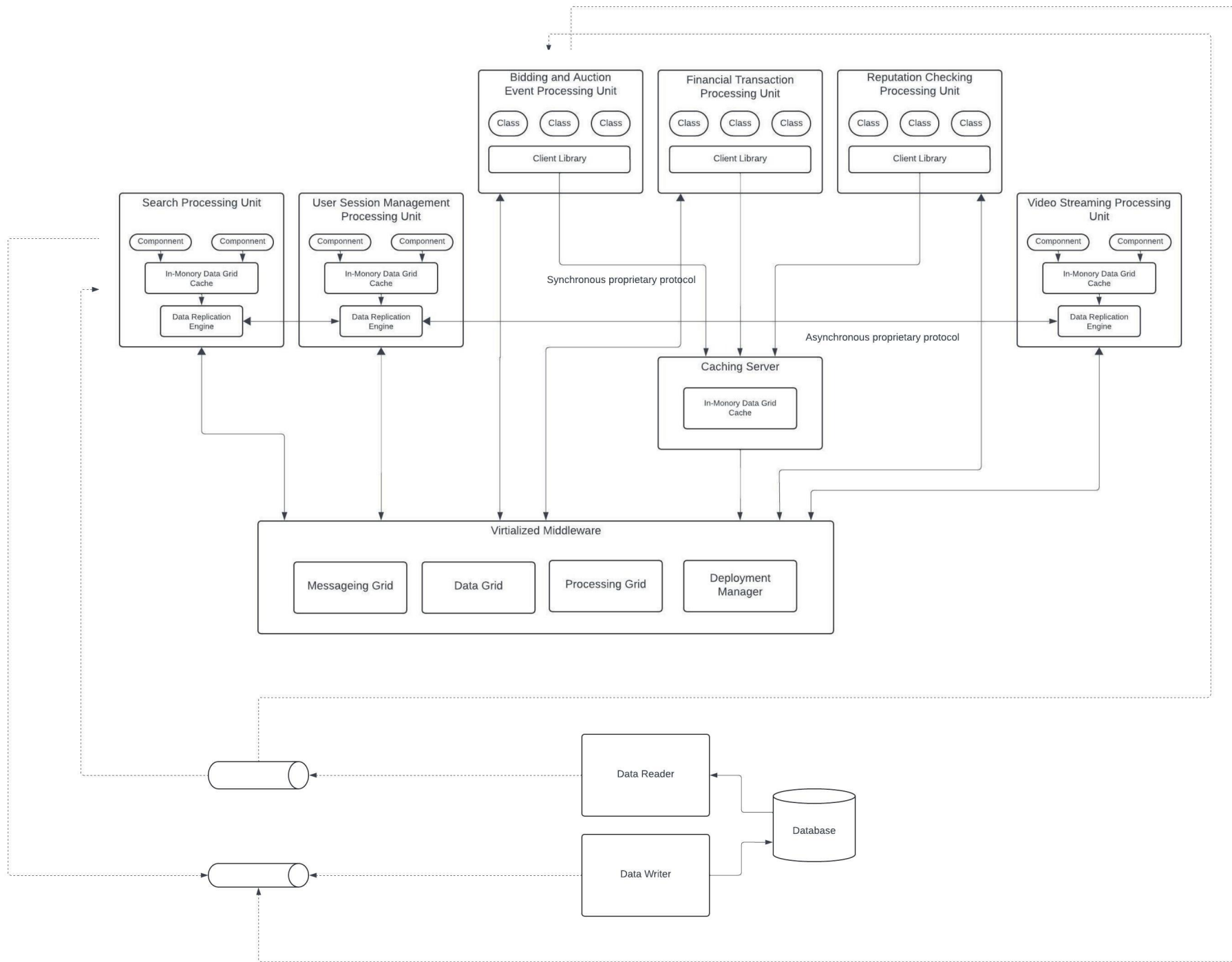
- The goal is to bring a traditional auction company to the next level by moving their operations online on a nationwide scale. This includes real-time bidding for users across the country, and the system allows users to participate in live and online auctions simultaneously. The system is expected to handle hundreds to thousands of participants per auction and to support multiple simultaneous auctions.
- The platform must handle money exchanges, support post-auction video streaming, and provide user user reputation checking. The company plans to expand overseas once the model is proven.

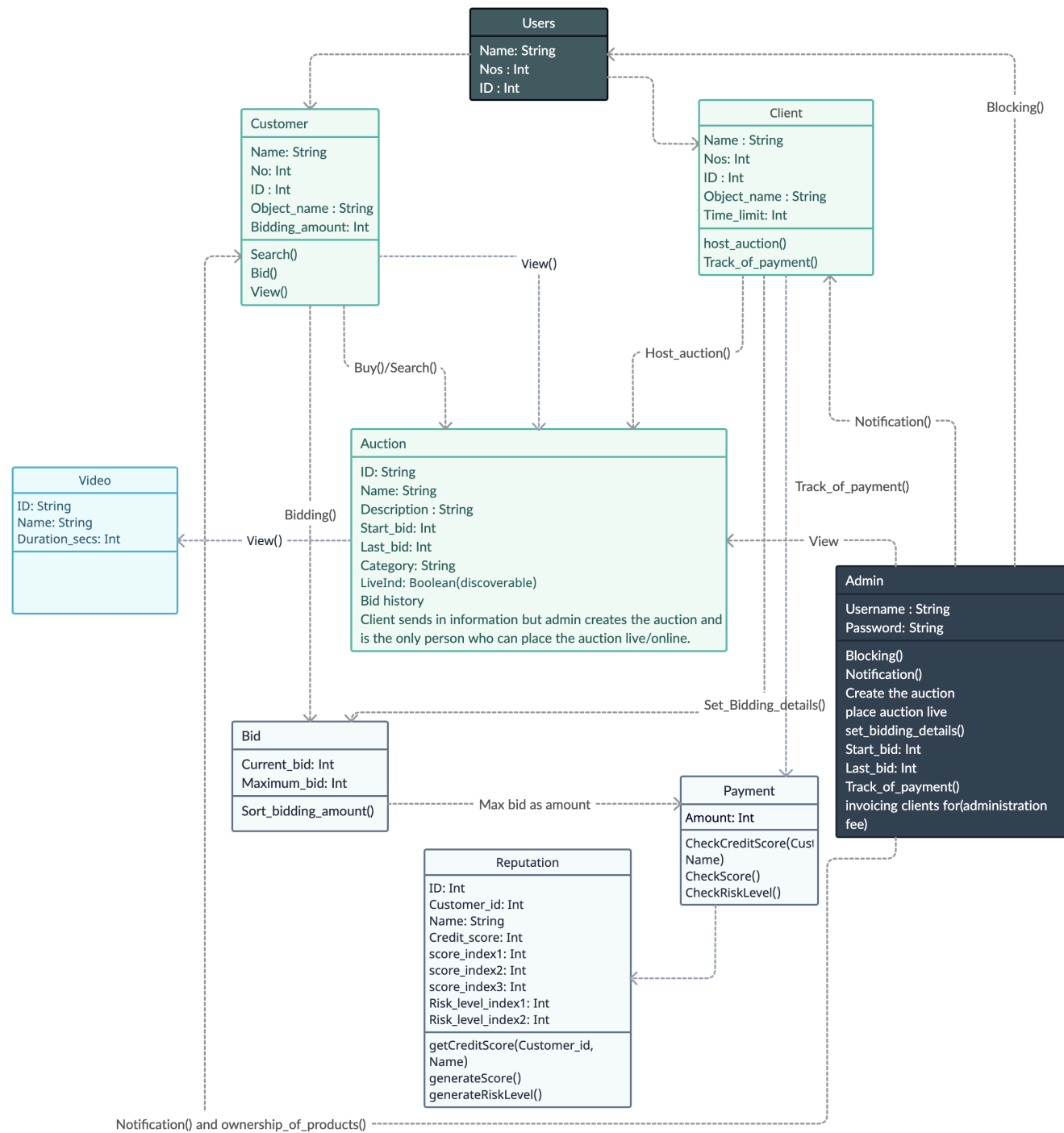
ARCHITECTURALLY INTERESTING BITS

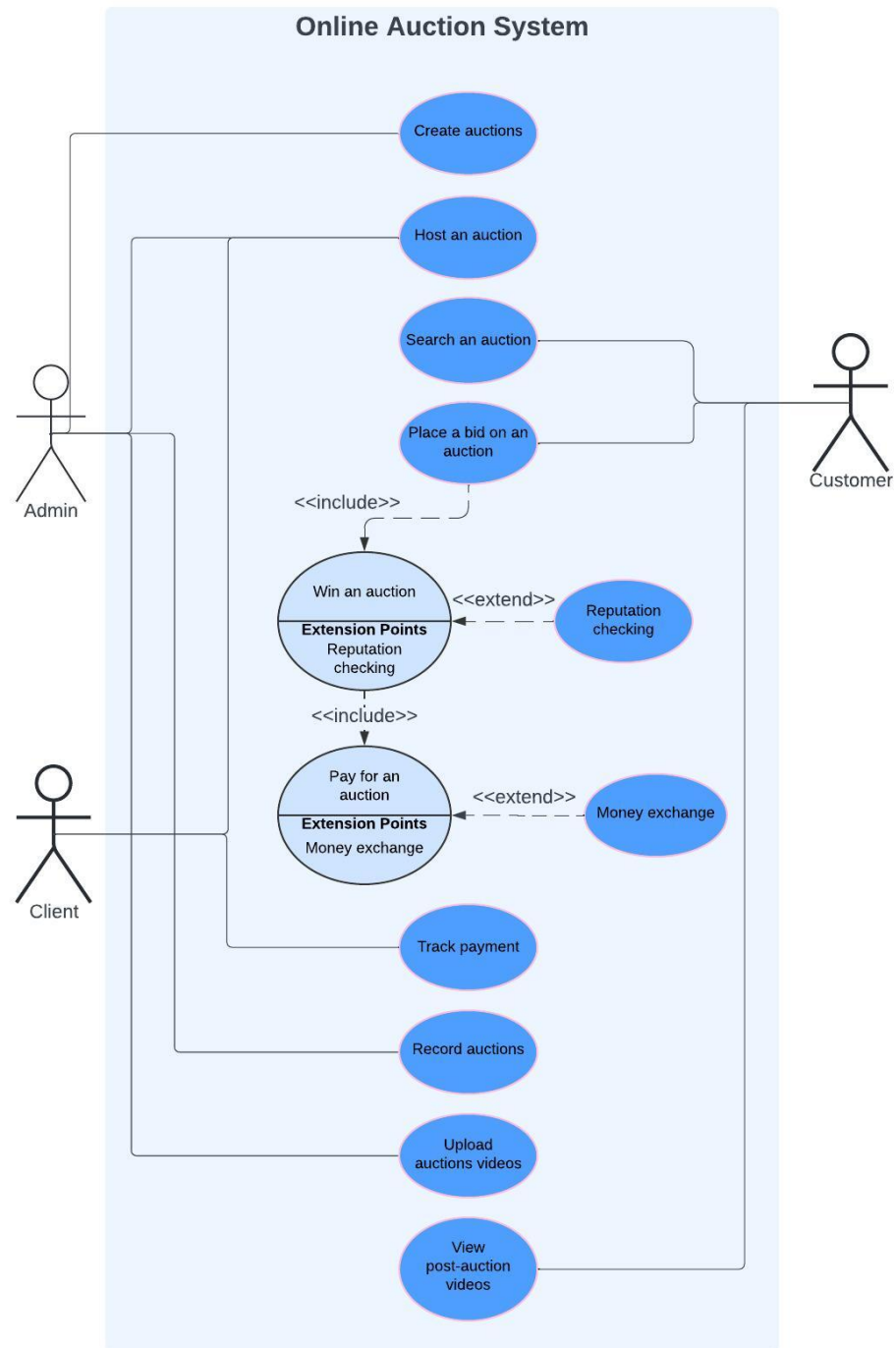
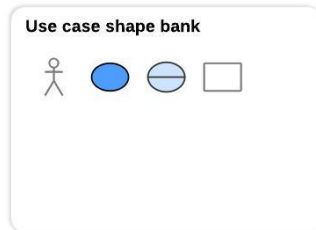
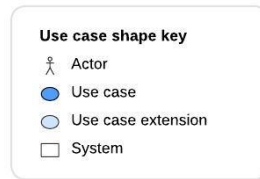
- **Real-time interactions:** Bidders expect their bids to be processed instantly.
- **Scalability:** The platform must handle many auctions and thousands of participants concurrently.
- **Fault tolerance:** The system must be resilient, with minimal downtime.
- **Data consistency:** Ensuring accurate, real-time updates on bids and user reputation.
- **Elasticity:** The platform must scale up or down based on auction load (users, bids, etc.).

PROPOSED SOLUTION

- To handle these challenges, I propose space-based architecture style. This architecture is ideal for scalable, high-traffic applications like an online auction platform.
- Multiple processing units can be started as the load increases; and as the auction winds down, unused processing units could be destroyed.
- Individual processing units can be devoted to each auction, ensuring consistency with bidding data.
- Due to the asynchronous nature of the data pumps, bidding data can be sent to other processing without much latency.







FAILURE CASES AND MITIGATIONS

- **System Overload During Peak Auction Times:** Automatic scaling to add more processing units as needed, which ensures the system can handle sudden increases in traffic.
 - Cloud platforms like AWS, Azure, OCI will automatically provision additional resources when thresholds (like high CPU usage) are exceeded.
- **Data Inconsistency in Bids:** Real-time bidding needs consistency; the highest bid must always be accurate across all users.
 - The central data grid will make sure data synchronization between the auction processing units, while eventual consistency will make sure updates are propagated even during minor delays.
- **Downtime During Auction Events:** Unscheduled downtime can result in lost revenue and unhappy customers.
 - The system will leverage a distributed architecture with failover capabilities. If part of the system fails, auctions will continue running without interruption. Backup services will ensure zero data loss.

BUSINESS DRIVERS AND AFFECTIONS

- **Scalability to International Markets:** The platform's architecture is designed to support horizontal scaling, and it has capacity to expand to new regions and countries seamlessly.
- **Customer Trust and User Experience:** The recent fraud lawsuit highlights the need for reputation management. The system will track participant behavior to ensure trustworthiness in future auctions, which is crucial for customer loyalty and growth.
- **Revenue Growth:** By handling many simultaneous auctions and thousands of participants reliably, the platform is expected to increase auction activity and revenue.
- **Compliance and Security:** Financial transactions must be secure and auditable to comply with global regulations because our future plans for overseas expansion.

EXPECTED DATA GROWTH AND IMPACTS

- **Auction Data:** As the company merges with smaller competitors and adds more users, the number of auctions, bids, and transactions will increase significantly. We estimate that the platform will handle millions of bids monthly as it scales nationally and internationally.
- **User Data and Reputation Scores:** Each new participant's activity (bids, transactions, ratings) will generate more data.
- **Media Video Data:** Once international expansion begins, more and more auction will be held and more and more post-auction video will be generated.
- **Impacts:** With increasing users and auction, data storage will need to expand. More replication must need. This increase the infrastructure cost and data grid scalability.

BUSINESS CONTINUITY NEEDS

- The platform must be operational 24/7, especially during peak auction hours. Zero downtime is the goal during auctions, and backup and disaster recovery strategies are critical.
 - **Hot Standby System:** If the primary system experiences issues, a secondary system (in another data center or region) will automatically take over to ensure uninterrupted service.
 - **Data Replication:** We'll use real-time data replication to ensure data consistency. If a data center fails, no auction data is lost, and the system can recover within minutes.